

Haniff Hamdan

Cloud / DevOps | AWS | Terraform | Docker & Kubernetes | CI/CD

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SUMMARY

Self-taught Cloud/DevOps engineer transitioning from full-stack development, with production experience on government enterprise systems (JPJ, Java/Spring Boot) and independent AWS delivery: Terraform/ECS/RDS environments, OIDC CI/CD, a serverless IoT pipeline, and a human-in-the-loop AI ops agent that diagnoses infrastructure anomalies with Gemini API and gates all execution behind Telegram approval — orchestrated via Google Antigravity 2.0. AWS re/Start graduate, certification in progress.

CLOUD & DEVOPS PROJECTS

AWS Infrastructure Provisioner — Terraform, ECS, RDS, CI/CD Personal Project

- Provisioned a production-style AWS environment from scratch with Terraform — VPC, ECS Fargate, RDS (MySQL on Graviton/ARM), IAM with GitHub OIDC federation — to deploy an existing Laravel application as a second, independent deployment target alongside its bare-metal Kubernetes (K3s) production environment.
- Built a GitHub Actions CI/CD pipeline authenticating to AWS via OIDC (no long-lived access keys), automatically building and pushing Docker images to ECR and triggering ECS deployments on every push.
- Implemented least-privilege IAM throughout: a dedicated Terraform service account scoped to only the permissions required, separate ECS execution/task roles, and application secrets pulled at runtime from SSM Parameter Store rather than baked into the container image.
- Diagnosed and resolved a multi-stage Docker build issue where frontend assets failed to compile inside the container, restructuring the Dockerfile into a two-stage build (Node.js asset compilation feeding a lightweight PHP runtime).

url: <http://54.169.236.18/>

Serverless IoT Smart Plug Pipeline — IoT Core, Lambda, DynamoDB Personal Project

- Built a serverless, event-driven pipeline ingesting real-time power consumption data from physical smart plugs via MQTT into AWS IoT Core, with Lambda enriching each reading against Malaysia's tiered residential electricity tariff (TNB Tariff A) before persisting to DynamoDB.
- Designed the pipeline VPC-free to eliminate idle infrastructure cost, using DynamoDB on-demand billing and TTL-based auto-expiry to keep storage cost near zero for a low-traffic personal deployment.
- Exposed processed data through a CORS-enabled API Gateway backed by a second Lambda, consumed by a dashboard hosted on S3 and served via CloudFront with Origin Access Control restricting direct bucket access.
- Resolved cross-environment integration issues spanning Python virtual environment isolation under PM2, module import ordering, and Terraform provider/lockfile conflicts when packaging Lambda functions.

url: <https://d30t4ra3ip2jtf.cloudfront.net/>

Human-in-the-Loop AI Ops Agent — Gemini, Prometheus, MQTT, Telegram Personal Project

- Built a persistent Python agent watching homelab (Prometheus) and IoT (MQTT smart plug) metrics, using Gemini 2.5 Flash to diagnose anomalies and propose remediation actions — with a human approval gate via Telegram inline buttons enforcing that no action ever executes without explicit sign-off.
- Engineered a three-layer safety architecture: a harness validating all LLM output before it reaches execution; a fixed action allowlist (PM2 restart, Docker service restart, K3s deployment scaling within a 1–3 replica clamp) as the sandbox boundary; and context engineering supplying a static system profile, a JSON-persisted incident history, and live metrics to every prompt call.
- Implemented smart plug idle detection via a local MQTT pipeline — bypassing AWS IoT for low-latency local alerting — with configurable wattage threshold and cooldown logic to avoid alert spam when a device remains idle over time.
- Applied a harness guardrail that downgrades any action returned outside the allowlist to a no-op before it reaches the executor — ensuring the LLM is strictly the diagnostic layer and never the trigger, even on malformed or hallucinated output.

Homelab Infrastructure — Self-Hosted Platform Engineering

Ongoing

- Built a Gemini 2.5 Flash-powered AI insights layer into the homelab dashboard that continuously reviews system metrics and logs, surfacing plain-language SRE summaries and anomaly explanations, with alerts pushed to Telegram
- Operate a self-hosted Kubernetes (K3s) cluster on ARM64 hardware (Orange Pi 4 Pro) running a full observability stack — Prometheus, Grafana, and Loki/Promtail for centralized log aggregation — replicating enterprise monitoring and log management workflows with LogQL-based error filtering..
- Designed and deployed HelmDeck, a self-hosted GitOps/Helm deployment manager (NestJS, React/TypeScript, PostgreSQL, BullMQ, Socket.io), framed as an Internal Developer Platform (IDP).
- Built and operate a production Laravel application (PHP 8.4-FPM, Nginx, Redis, MariaDB) with enterprise patterns including repository pattern, soft deletes, DB transactions, and response caching, accessed publicly via a GCP reverse-proxy entry point over Tailscale.
- Resolved infrastructure-level networking issues independently, including K3s iptables chain conflicts blocking inter-container traffic and CSRF failures through a reverse proxy.
- Samba Protocol for media and cloud storage

EXPERIENCE

Full-Stack Developer

09/2023 – Current

Heitech Padu Bhd

Project: JPJ Teknologi Digital

- Developed and maintained backend services using Java (Spring Boot) to support vehicle registration and licensing systems
- Optimized DB2 database queries, improving processing efficiency and reducing data inconsistencies
- Built and executed integration tests to validate API functionality and ensure system reliability
- Collaborated in Agile sprints, contributing to backlog refinement, sprint planning, and cross-team delivery

Project: KPDRM System Revamp

- Redesigned and modernized legacy system architecture using C# (.NET) and REST APIs
- Improved SQL query performance, reducing data retrieval time by approximately 30%
- Developed responsive frontend components aligned with backend architecture for improved UX
- Led UAT testing cycles, identifying critical issues and reducing production defects by 20%

System Developer

06/2023 – 09/2023

NIISE, IRIS Corporation Bhd

- Engineered backend APIs using C# and ASP.NET to process biometric data from fingerprint scanners
- Integrated hardware authentication systems with web applications for secure real-time verification
- Enhanced API performance to support high-speed, secure biometric data transmission
- Re-engineered immigration system modules, improving system functionality and scalability

Software Developer Intern

07/2022 – 01/2023

Bravonet Sdn Bhd

- Developed responsive web interfaces using JavaScript, Bootstrap, and CSS with cross-browser compatibility
- Collaborated with senior developers to debug and resolve UI issues, improving application performance
- Participated in Agile ceremonies including sprint planning, standups, and retrospectives
- Supported CI/CD processes by assisting in build, testing, and deployment workflows

SKILLS

Cloud & Infrastructure: AWS (VPC, ECS Fargate, RDS, IAM, IoT Core, Lambda, DynamoDB, S3, CloudFront, API Gateway, SSM Parameter Store)

Infrastructure as Code: Terraform (state management, remote backends, modules)

Containers & Orchestration: Docker, Kubernetes (K3s), Helm

CI/CD & Version Control: GitHub Actions (OIDC, environment protection rules), Git

Observability: Prometheus, Grafana, Loki/Promtail, CloudWatch

Linux & Networking: Linux administration, NGINX, MQTT, Tailscale

Backend Development: Java (Spring Boot), C# (.NET/ASP.NET), Node.js (NestJS), PHP (Laravel)

Frontend Development: React.js, Angular, TypeScript

Databases: MySQL/MariaDB, PostgreSQL, DynamoDB, SQL optimization

Practices: Agile/Scrum, Test-Driven Development, Microservices Architecture

EDUCATION

Bachelor of Engineering, Technology Electronic (Computer System) 2019 – 2023
Universiti Malaysia Pahang

Diploma, Electronic Engineering (Communication System) 2015 – 2018
Politeknik Kota Bharu

CERTIFICATIONS & LICENSES

- DevOps Foundation Certification
- Advanced LookML Concepts (Looker)
- JavaScript Algorithms and Data Structures
- Responsive Web Design • Wireframing and Prototyping • Foundations of UX Design

LEADERSHIP & PROGRAM EXPERIENCE

AIESEC — Director, Outgoing Global Volunteer: Led team operations and stakeholder engagement; managed international partnerships

PEKA 4 — Discipline & Safety EXCO: Coordinated safety initiatives and compliance awareness programs